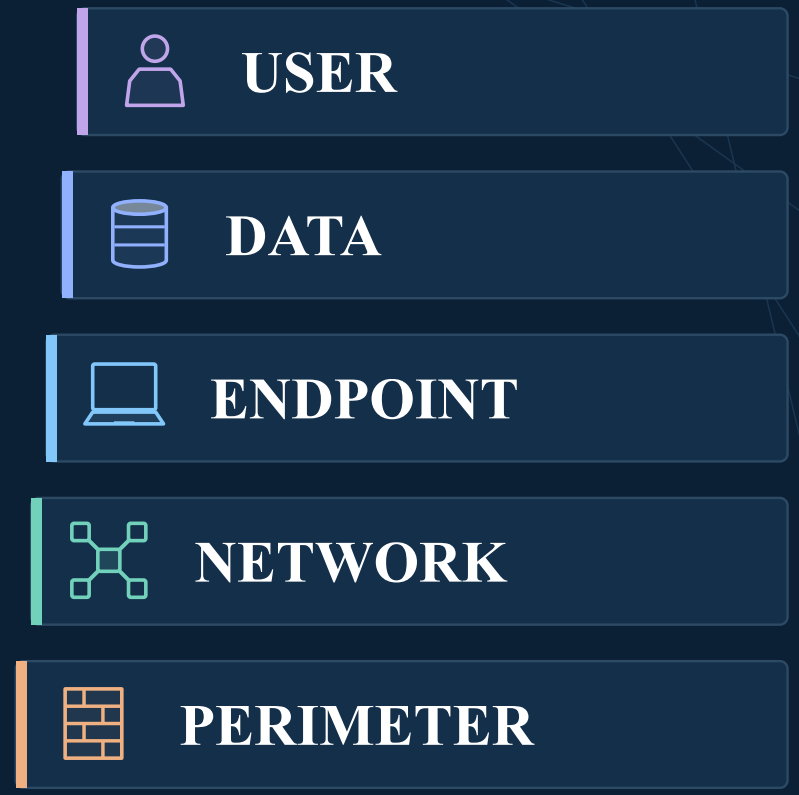
Network Security

All Layers Security Concept

DSI102 — Introduction to Digital Technology | Week 8

Dr. Nantapong Keandoungchun



This week: build the defenses.
Next week: study how cyber attacks try to break them.

How many layers stand between a user and valuable data?

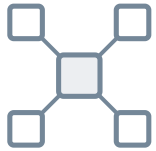
A secure system is not one wall. It is a chain of coordinated protections.



Today we study the protective layers — not the attack techniques.

Where Week 8 fits in the course

From communication → protection → attacks.



Network Communication

How devices exchange data



Week 8: Security Layers

How systems are protected



Next Week: Cyber Attacks

How attackers try to bypass defenses

> Learning sequence: understand the system first, then understand how it can be attacked.

Learning outcomes

By the end of Week 8, students should be able to…



Explain layered security

Why multiple controls are used together.



Identify each security layer

Perimeter, network, endpoint, application, data, user, cloud.



Map controls to layers

Firewall, VPN, NAC, patching, ACL, backup, IAM, etc.

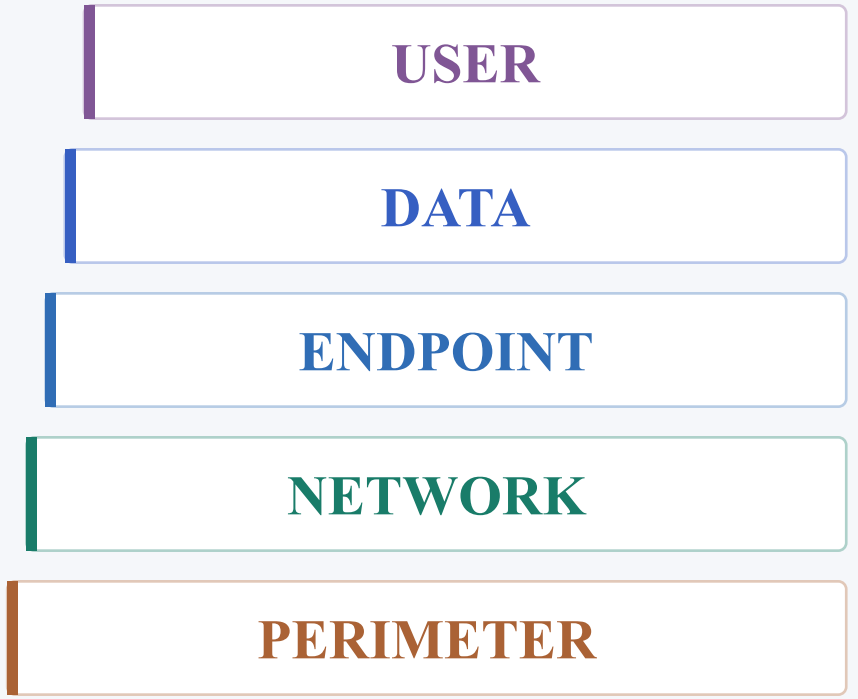


Explain Defense in Depth

How layers combine across prevent, detect, respond, recover.

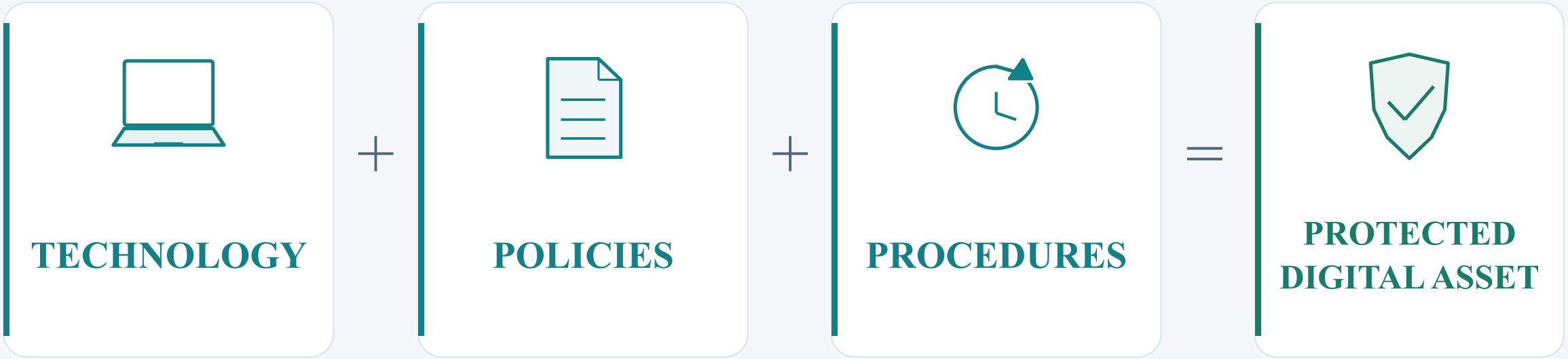
Today’s learning journey

1. **Layered Security Concept**
2. Perimeter & Network
3. Endpoint & Application
4. Data & User
5. Cloud & Workload
6. Defense in Depth



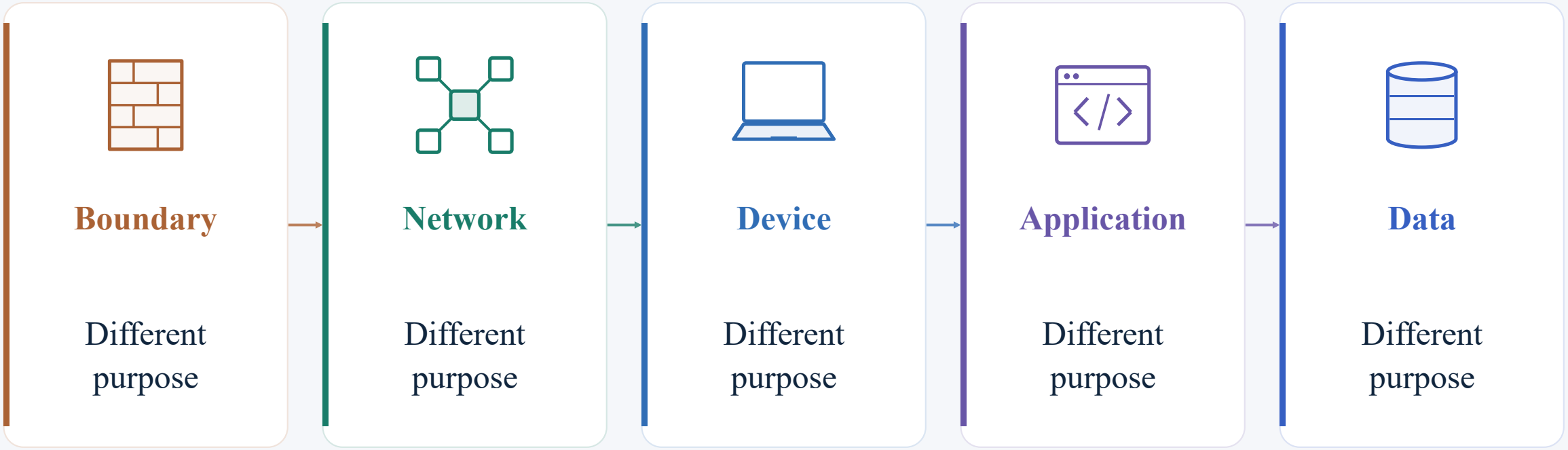
Layered Security

Use multiple layers of defense to protect digital assets.



> Layered security reduces dependence on any single control.

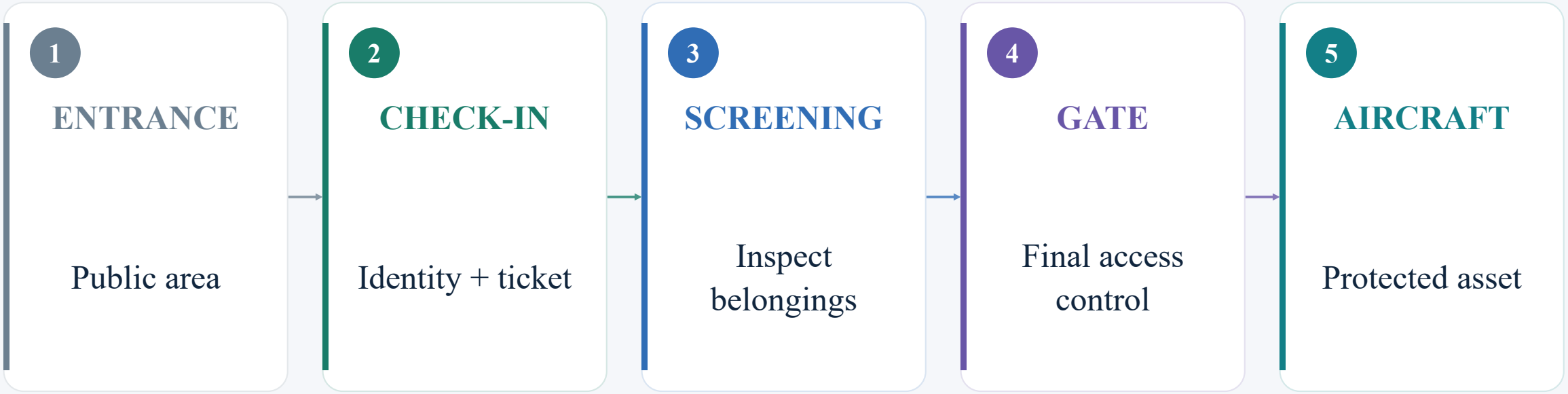
Why one security control is not enough



If every layer has a different job, failure in one does not automatically mean failure everywhere.

Real-life analogy: airport security

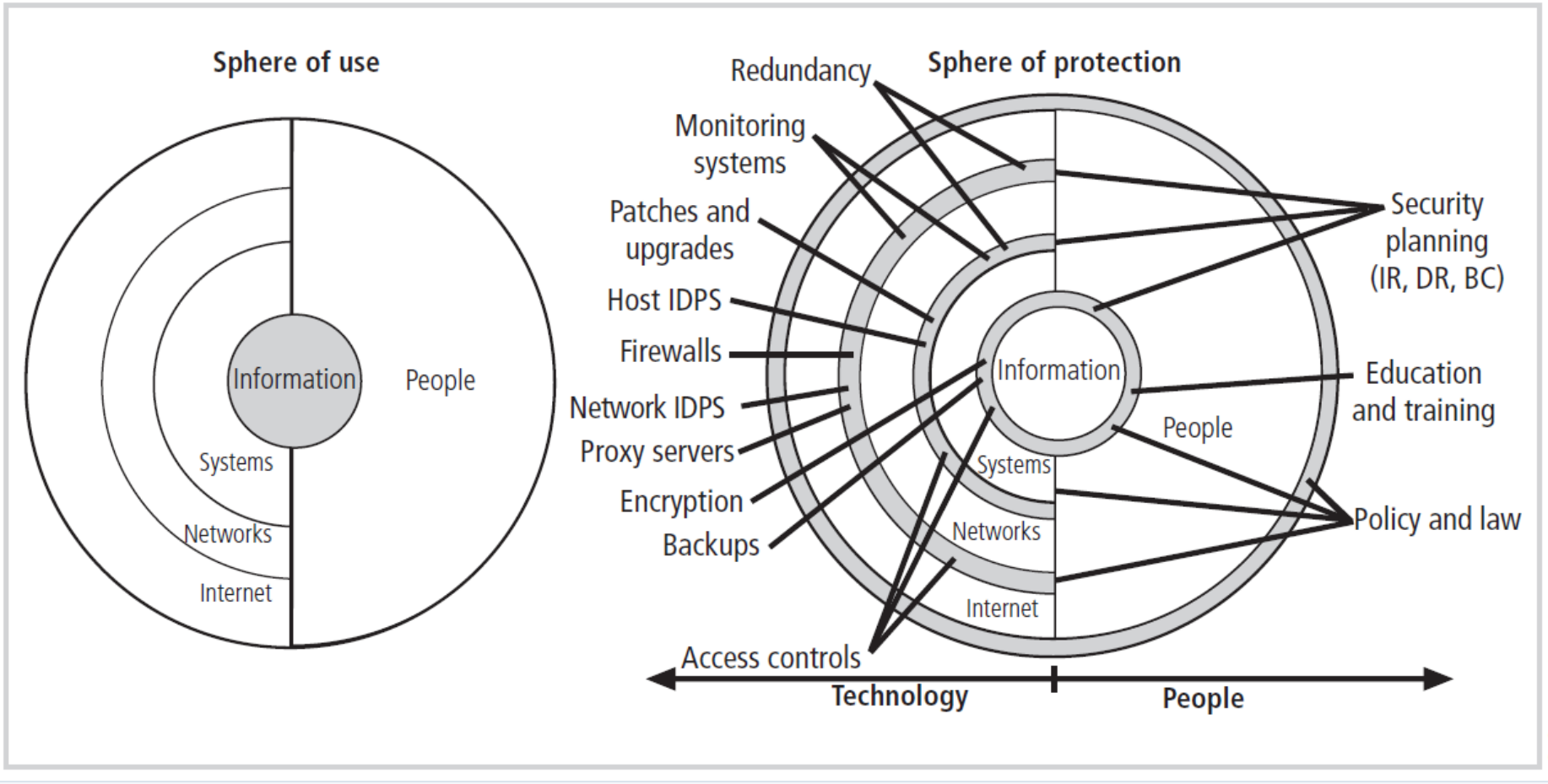
Layered security already exists in the physical world.



No single checkpoint does everything. Cybersecurity follows the same logic.

Spheres of Security

Information sits inside systems, networks, external connections, and people.



Five core layers of a security strategy

Week 8 covers the complete model.



Supporting layers: application security, cloud security, workload security.

Defense in Depth — preview

Layered security becomes stronger when controls are deliberately integrated.



PREVENT

Reduce the chance of compromise



DETECT

Notice unusual or harmful activity



RESPOND

Contain and manage an incident



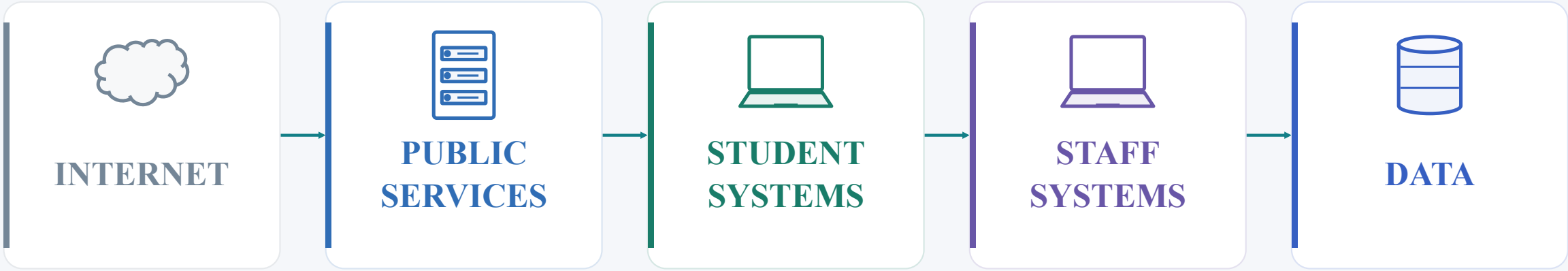
RECOVER

Restore services and data

We return to this framework after learning every layer.

Recurring example: a university digital environment

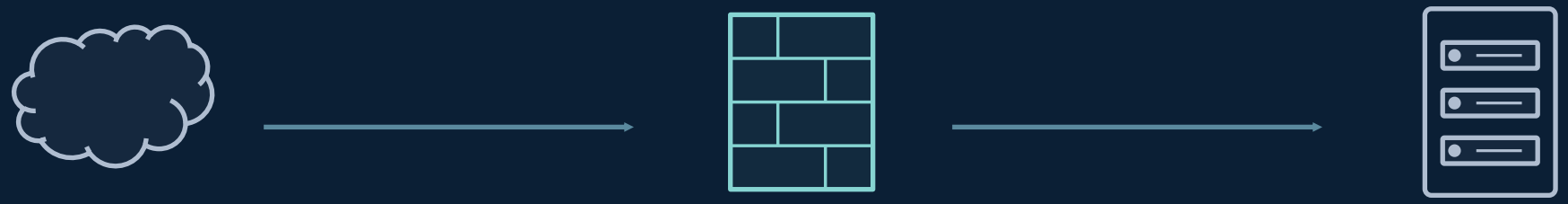
We will add security layer by layer.



> Question: where should each security control be placed?

First question

Where should the first defense be placed?



UNTRUSTED **BOUNDARY** **INTERNAL**

01

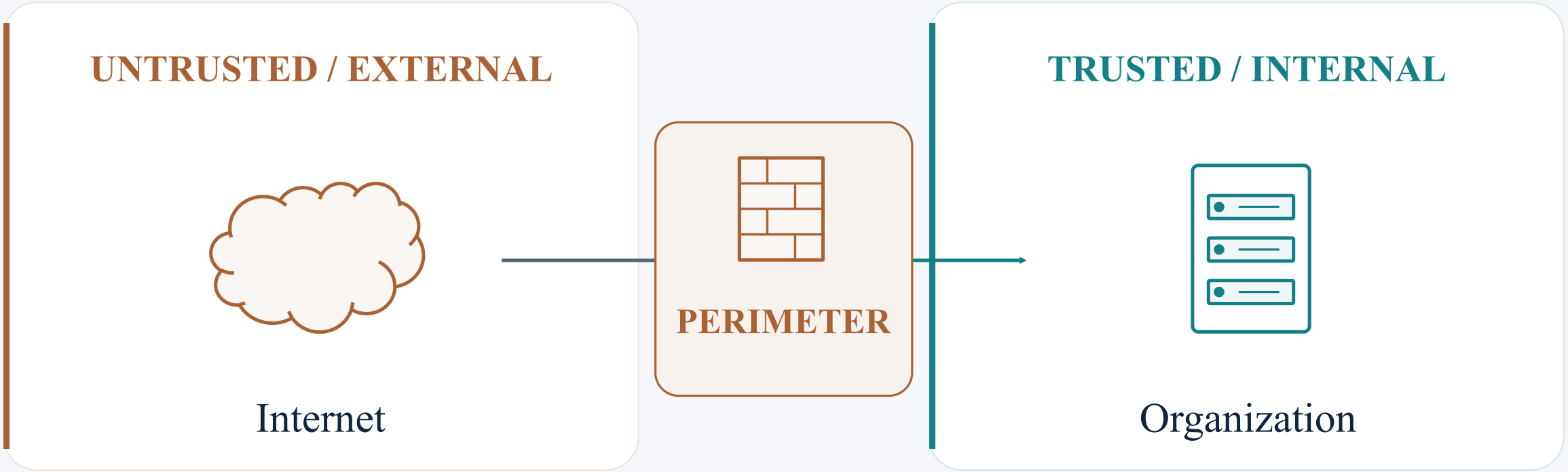
PERIMETER SECURITY

The outer defensive boundary.



Perimeter security

Protect the boundary between trusted and untrusted networks.



Firewall

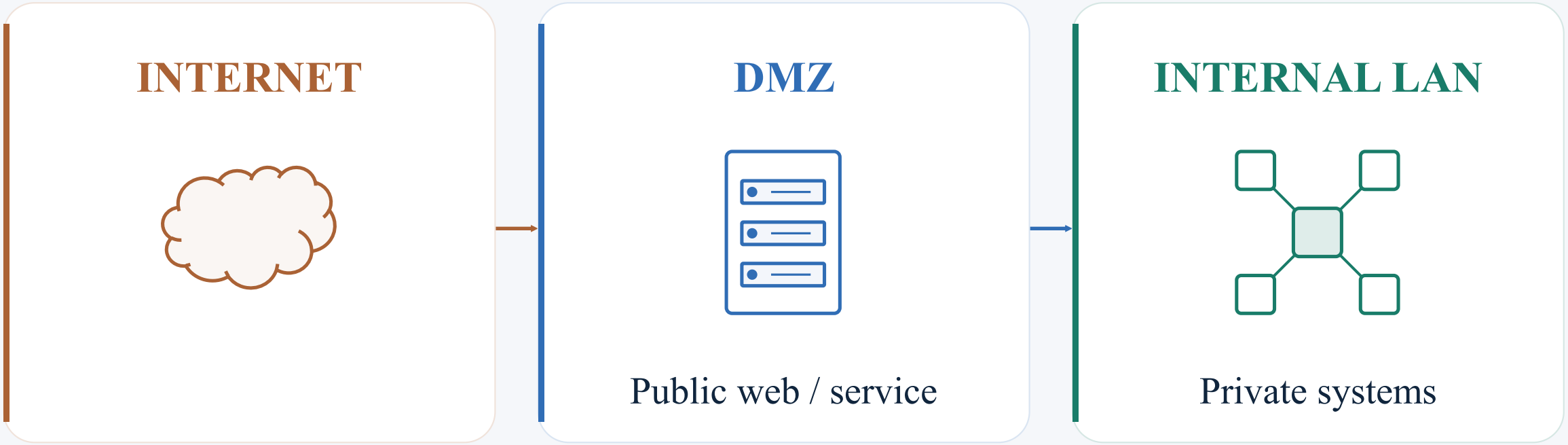
Traffic control at a security boundary.



Decision is based on defined security rules.

De-Militarized Zone (DMZ)

Separate public-facing services from the internal LAN.

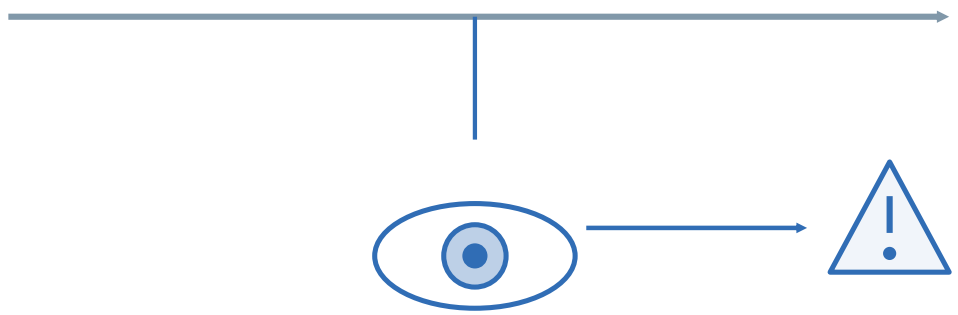


> The DMZ creates separation, not complete trust.

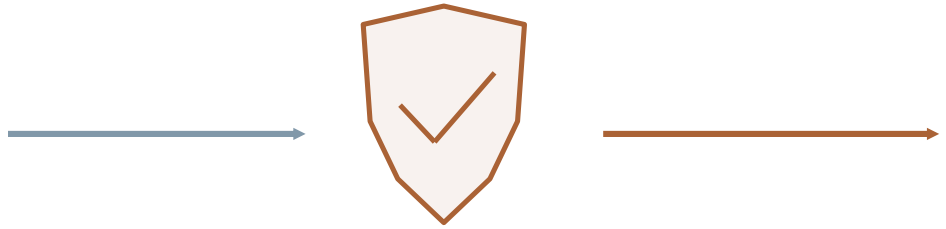
IDS vs IPS

Same goal: identify suspicious activity. Different action: alert vs intervene.

IDS — DETECT & ALERT



IPS — DETECT & PREVENT



Week 8 focus: understand their defensive roles, not attack signatures or evasion techniques.

Perimeter security — putting the controls together



Perimeter security manages exposure at the outer edge.

Check your understanding

Which control best matches each role?

Controls boundary traffic	**Firewall**
Separates public service from internal LAN	**DMZ**
Monitors and alerts	**IDS**
Can intervene in traffic	**IPS**

02

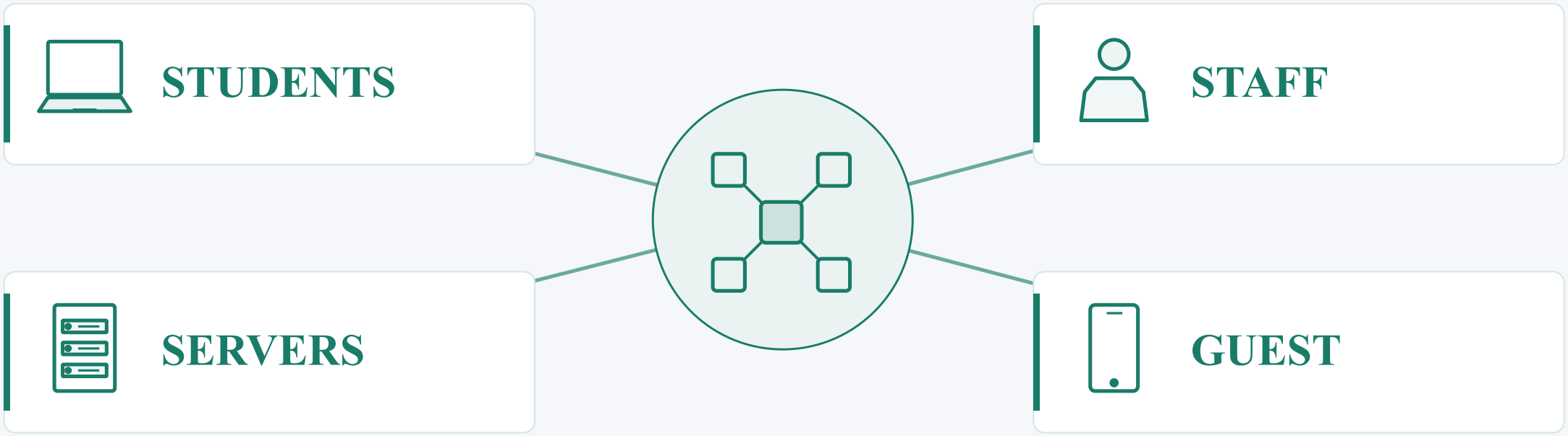
NETWORK SECURITY

Defense within the walls.



Network security

Protect internal communication and limit unnecessary movement.

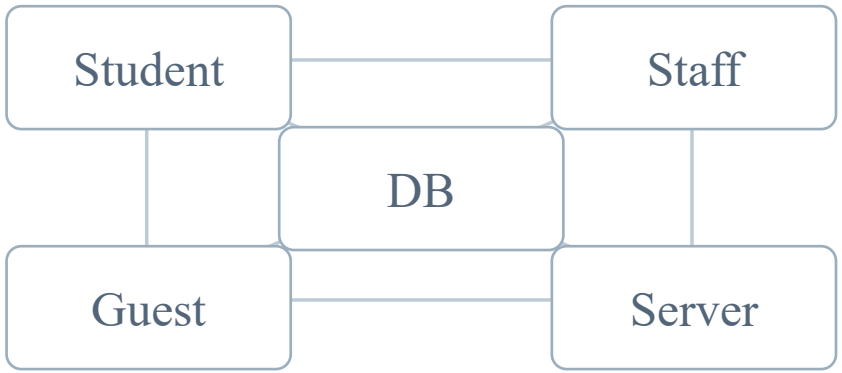


Goal: separate, monitor, and control internal paths.

Network segmentation

Divide a large network into logical or functional zones.

FLAT NETWORK



SEGMENTED NETWORK

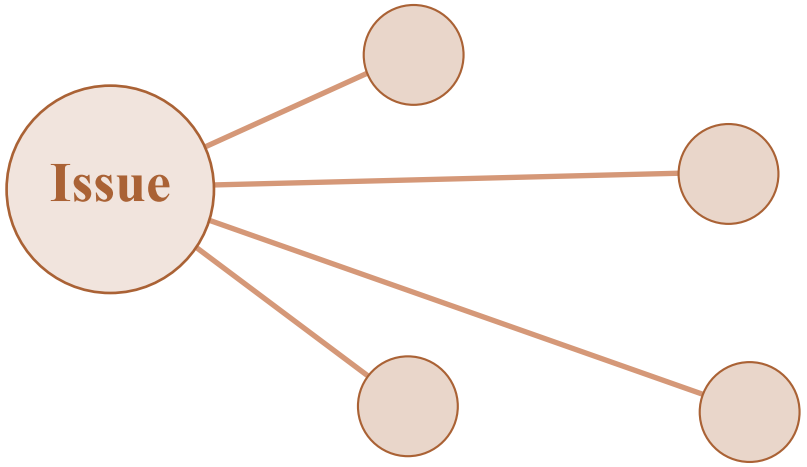
- STUDENT
- STAFF
- SERVER
- GUEST

Segmentation makes security policy easier to enforce and limits unnecessary reachability.

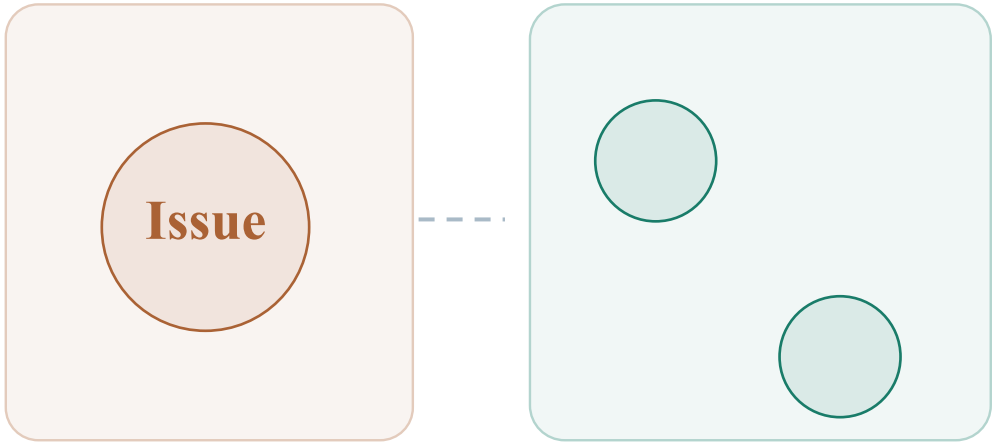
Segmentation changes the blast radius

Same incident, different architecture.

BEFORE: ONE LARGE NETWORK



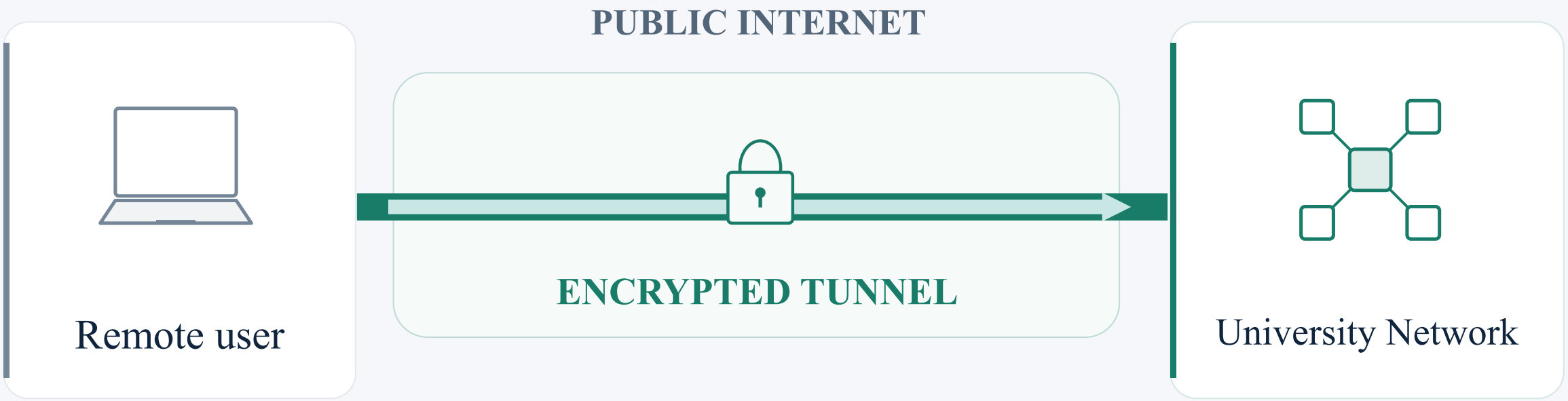
AFTER: CONTROLLED ZONES



> Containment is a design property, not an emergency trick.

Virtual Private Network (VPN)

Protect a connection from an endpoint to a network across the Internet.



Concept focus: protected communication across an untrusted network.

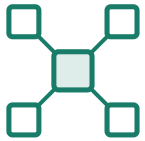
Network Access Control (NAC)

Recognize users and devices, then apply access policy.

1	STUDENT LAPTOP	NAC CHECK	Student network
2	STAFF LAPTOP	NAC CHECK	Staff services
3	GUEST PHONE	NAC CHECK	Internet only
4	UNKNOWN DEVICE	NAC CHECK	Restricted / denied

Network security — key idea

Inside the boundary, control movement and access.



SEGMENT

Separate zones and functions



VPN

Protect remote connections



NAC

Control who or what may connect



MONITOR

Observe internal traffic and events

Transition question

The network is protected — but what if the device itself is weak?



> Network security cannot replace endpoint security.

03

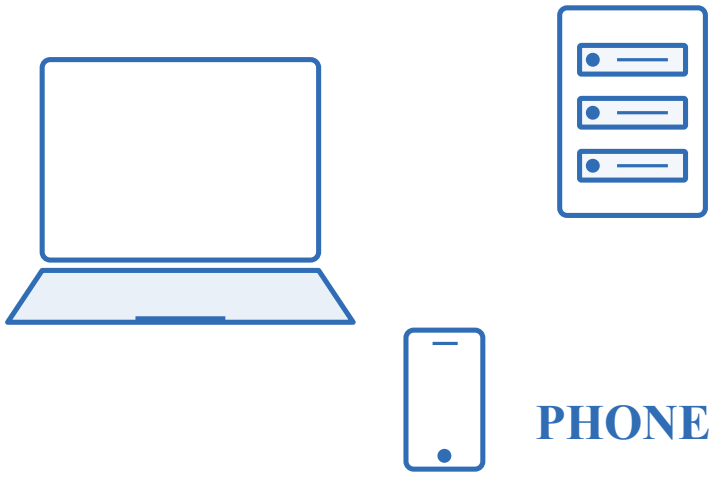
ENDPOINT SECURITY

Protect individual devices.



Endpoint security

Protect laptops, desktops, servers, and mobile devices.



- Anti-malware
- Patching
- Host monitoring
- Access control

Endpoint controls live where users and applications actually run.

Anti-virus and anti-malware

Scan, monitor, isolate, and remediate malicious software.

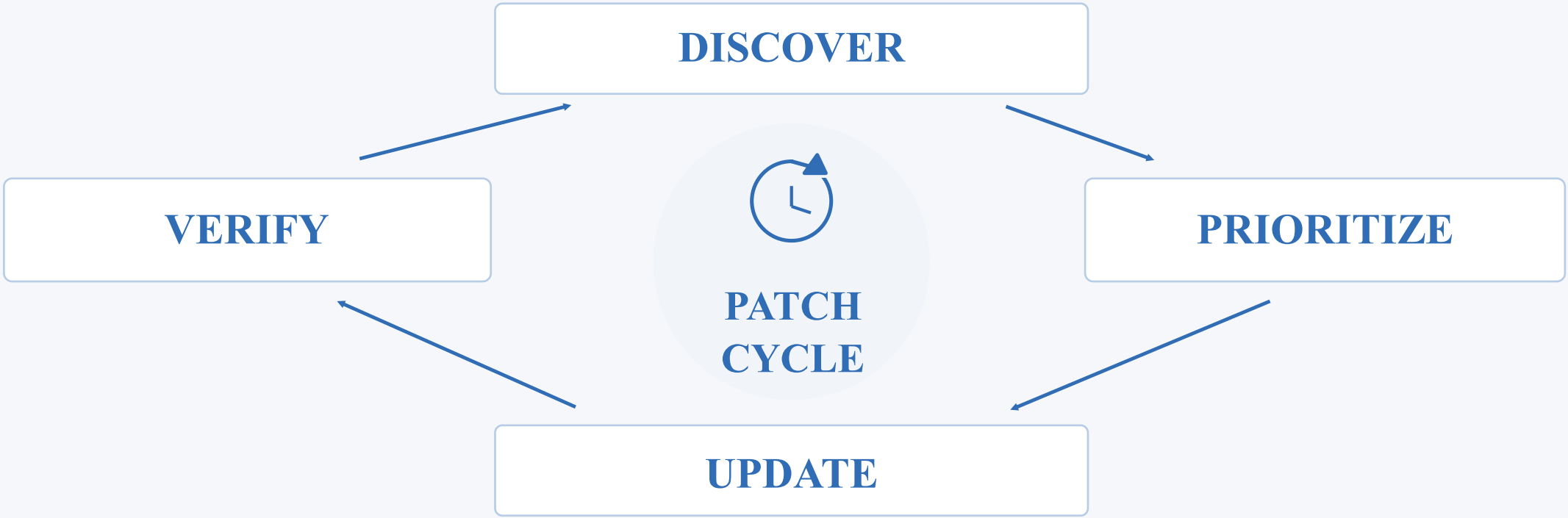


Continuous monitoring matters because suspicious behavior may appear after initial entry.

> This week: defensive concept only. Attack types and malware behavior are next week.

Patch management

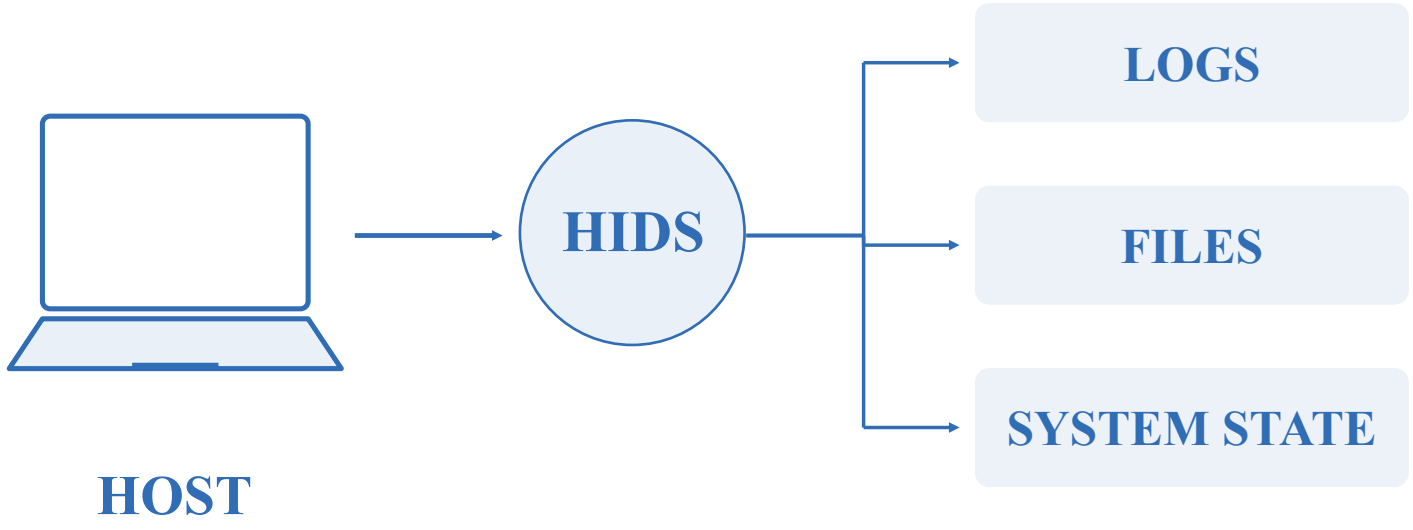
Fix known software weaknesses before they become larger risks.



> A secure endpoint is maintained, not installed once and forgotten.

Host-based monitoring

Watch what happens on an individual device.



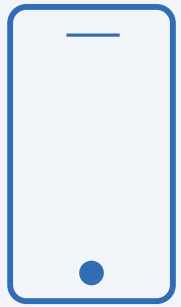
HOST VIEW

Network monitoring sees the road. Host monitoring sees what happens inside the building.

Mobile device security

Apply endpoint principles to smartphones, tablets, and wearables.

- ACCESS CONTROL
- DATA ENCRYPTION
- REGULAR UPDATES



- APP PERMISSIONS
- SECURE INSTALLATION
- ANTIMALWARE

Endpoint security — key idea



PROTECT

Block or isolate known risks



MAINTAIN

Patch and update software



MONITOR

Observe host behavior



CONTROL

Limit user/device access

> Devices are a security layer because they execute code, store data, and connect to networks.

04

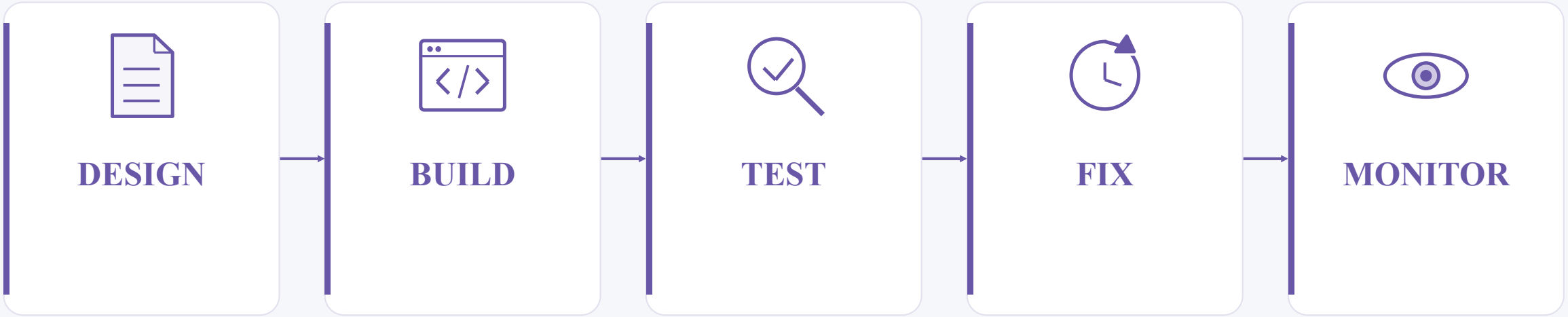
APPLICATION SECURITY

Protect the software that delivers digital services.



Application security lifecycle

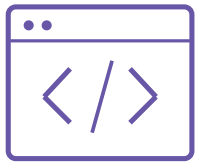
Find, fix, and prevent vulnerabilities throughout the application lifecycle.



> Security is part of the application process — not a final decoration.

Web security

Protect websites, web applications, and web services.



INPUT VALIDATION

Check input before processing it.

SESSION MANAGEMENT

Manage active sessions safely.

AUTHENTICATION

Verify who the user is.

AUTHORIZATION

Control what the user may do.

SENSITIVE DATA

Reduce unauthorized exposure.

Authentication, authorization, input, and session

Four different questions inside a secure web application.

1	WHO ARE YOU?	→ Authentication
2	WHAT MAY YOU DO?	→ Authorization
3	IS THIS INPUT SAFE TO USE?	→ Input validation
4	IS THIS ACTIVE SESSION STILL TRUSTED?	→ Session management

Email security

Filter incoming threats and control risky outbound messages.



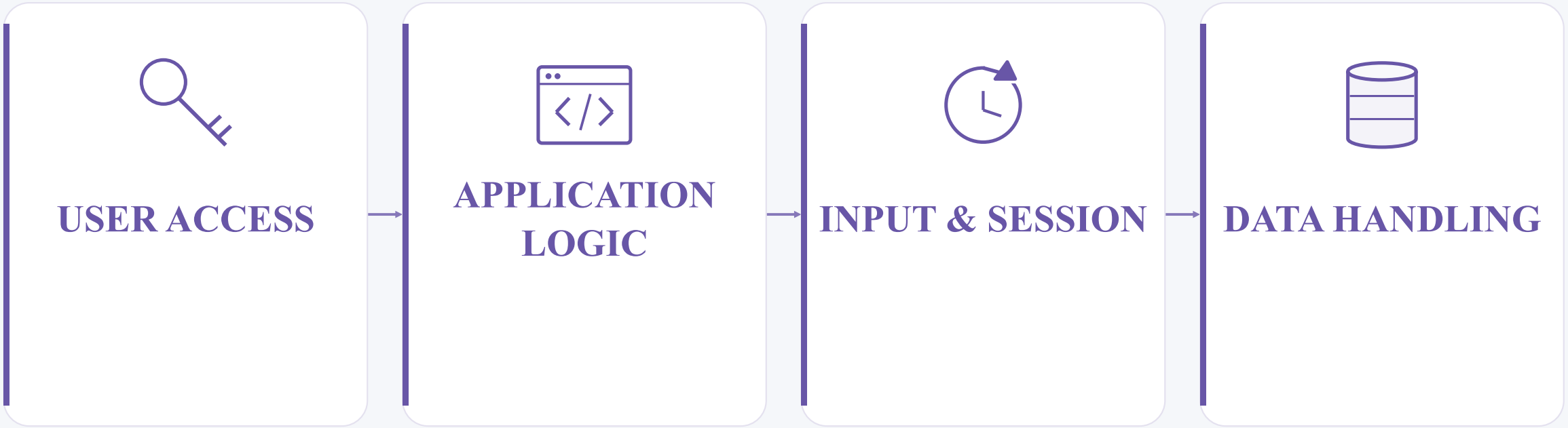
PHISHING

MALICIOUS CONTENT

SENSITIVE DATA LOSS

> Attack techniques are next week; today we focus on the defensive layer.

Application security — key idea



> Application security protects how software receives, processes, and exposes information.

05

DATA SECURITY

Protect the information itself.



Data security

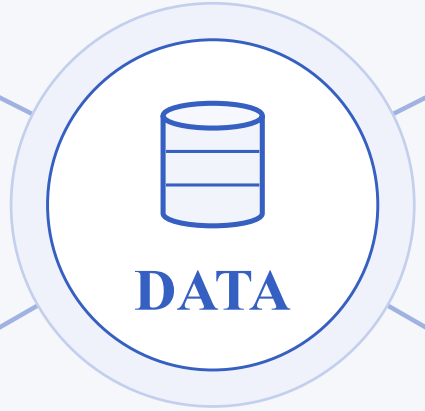
Protect access, confidentiality, availability, and recovery.

ACCESS CONTROL

Who may view or modify?

ENCRYPTION

How is data protected?



BACKUP & RECOVERY

How do we restore it?

DLP

How do we prevent unsafe loss?

Access control: who can do what?

Access Control Lists and matrices make permissions visible.

	FILES	GRADES	FINANCE	SYSTEM
STUDENT	R	-	-	-
LECTURER	R/W	R/W	-	-
ADMIN	R/W	R/W	R/W	ADMIN



PRINCIPLE

Permissions should reflect organizational roles and actual business needs.

Encryption

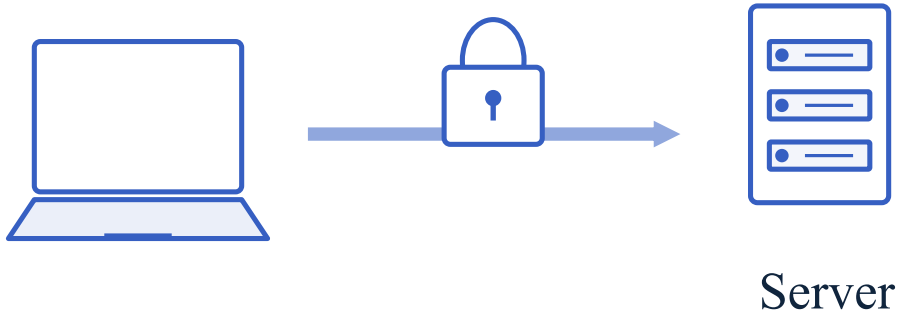
Protect data in storage and during communication.

DATA AT REST



Stored files, databases, backups

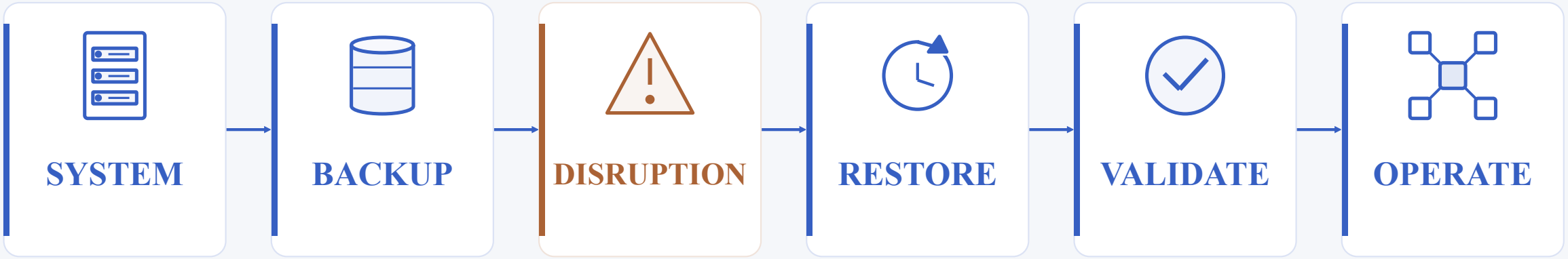
DATA IN TRANSIT



Protected communication path

Backup & recovery

Backups create a path back from loss or corruption.



FULL BACKUP

Complete copy; more time and storage.

INCREMENTAL BACKUP

Only changed files; smaller and faster.

Recovery objectives

Different metrics answer different recovery questions.

MTD

Maximum Tolerable Downtime

How long can the process be disrupted?

RTO

Recovery Time Objective

How fast must a system be restored?

RPO

Recovery Point Objective

How much data loss in time is acceptable?

WRT

Work Recovery Time

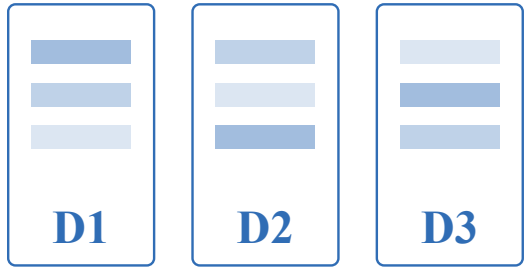
How long to validate and resume operations?

RAID — resilience at the storage layer

Redundancy can improve availability, but RAID is not a replacement for backup.

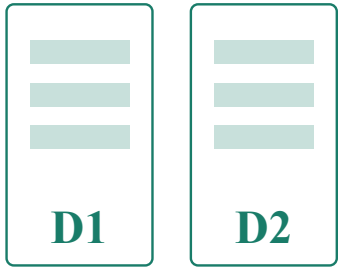
RAID 0

Striping



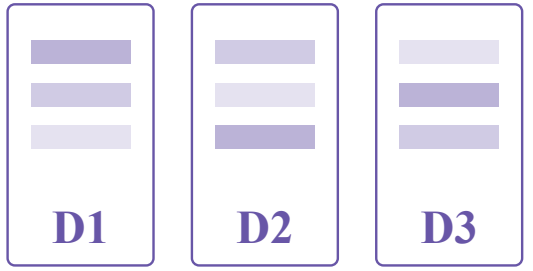
RAID 1

Mirroring



RAID 5

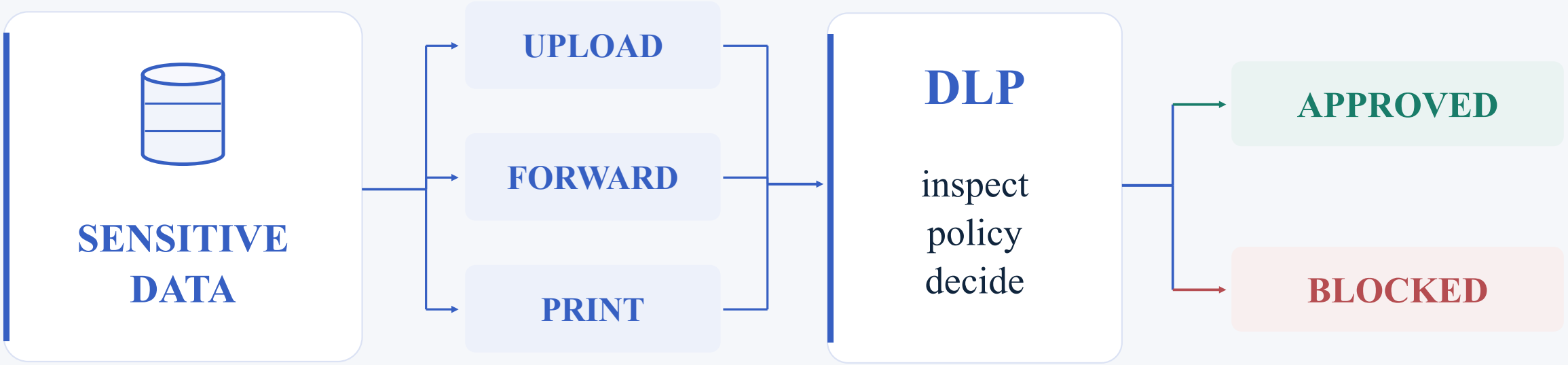
Striping + parity



> Availability technique ≠ recovery strategy.

Data Loss Prevention (DLP)

Apply policy before sensitive information leaves the organization.



Data security — key idea



CONTROL ACCESS

Limit who can view or change



PROTECT CONTENT

Encrypt stored and transmitted data



PREVENT LOSS

DLP and policy



RECOVER

Backup, recovery objectives, redundancy

06

USER AWARENESS & POLICY

People are part of the security system.



User awareness

Technical controls cannot compensate for every unsafe decision.



RECOGNIZE RED FLAGS

Suspicious email, website, request



FOLLOW PASSWORD RULES

Use approved password practices



USE APPROVED SERVICES

Avoid unapproved tools and workarounds



REPORT PROBLEMS

Escalate suspicious events quickly

Focus on behavior, not only knowledge.

Information security policies and procedures

Turn security expectations into repeatable organizational practice.

DEVELOP POLICY

↓

ACCESS CONTROL & AUTHENTICATION

↓

PASSWORD POLICY

↓

SECURITY AWARENESS TRAINING

WHY?

Policies connect people, access decisions, and training into one operating model.

Identity protection: password + MFA

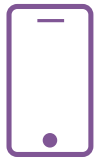
Use more than one proof of identity when appropriate.



PASSWORD

Something you know

+



2nd FACTOR

Something you have / are

=



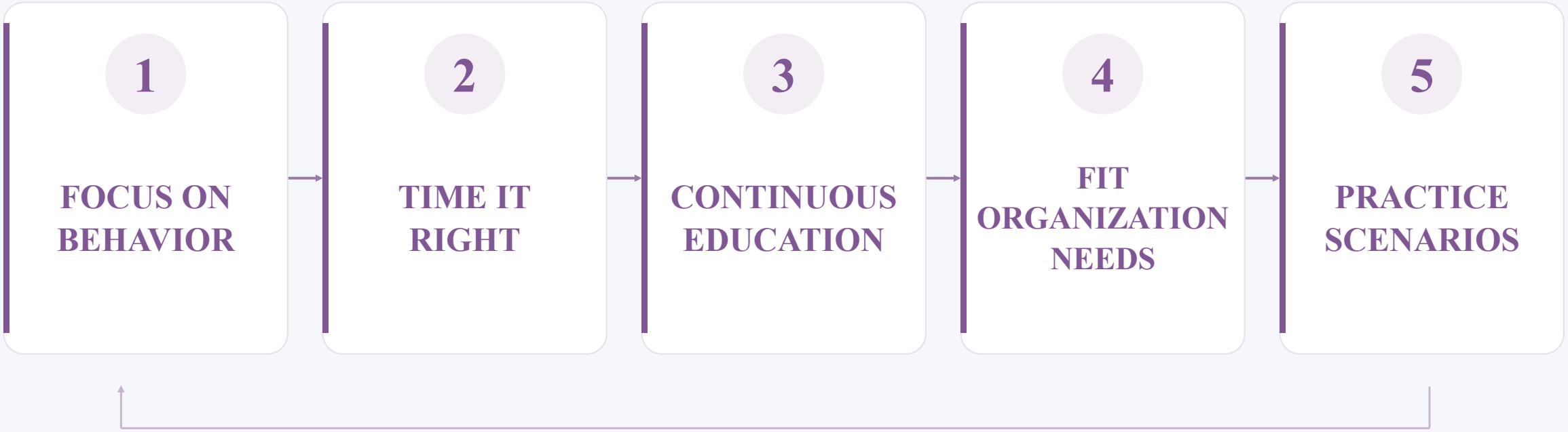
STRONGER IDENTITY CHECK

POLICY

Strong identity practices belong to the user layer and also support cloud/application access.

Security awareness should be continuous

Build behavior through repeated, relevant learning.



> Training is a control that changes human behavior over time.

The human layer — key idea

AWARENESS

Recognize warning signs

POLICY

Know expected behavior



IDENTITY

Use strong authentication

REPORTING

Escalate suspicious activity

People can strengthen or weaken every technical layer.

07

CLOUD & WORKLOAD SECURITY

Extend the same principles into distributed environments.



Cloud security

Protect data, applications, services, and infrastructure in cloud environments.



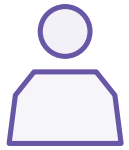
Shared responsibility model

Cloud security is divided between the provider and the customer.



CLOUD PROVIDER

Protects the underlying cloud service and associated infrastructure according to the service model.



CUSTOMER

Configures identities, access, applications, data protection, and other controls that remain under customer responsibility.

SHARED The exact split depends on the cloud service model.

Identity and Access Management (IAM)

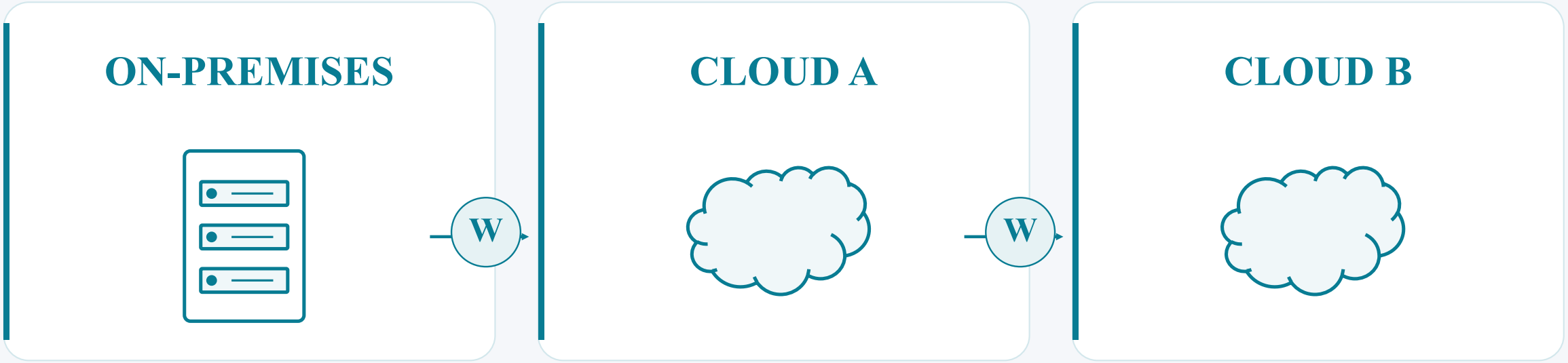
Control identities and permissions in cloud services.



> Identity is often the front door to cloud services.

Workload security

Protect workloads as they move across cloud and hybrid environments.



SECURITY CHALLENGE

Distributed workloads increase the attack surface and must be secured without removing the agility of hybrid environments.

Cloud network controls

Cloud environments still need traffic boundaries and access policy.



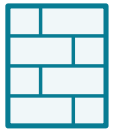
SECURITY GROUPS

Instance-level traffic rules



ACLs

Subnet / network access rules



FIREWALL / IDPS

Inspect and monitor traffic



ENCRYPTION

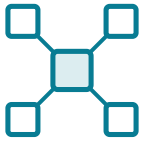
Protect data paths and storage

Cloud security — key idea



IDENTITY

Control who can access



NETWORK

Control cloud traffic



DATA

Protect cloud information



WORKLOAD

Protect distributed services

> Cloud security reuses the same principles in a different operating environment.

08

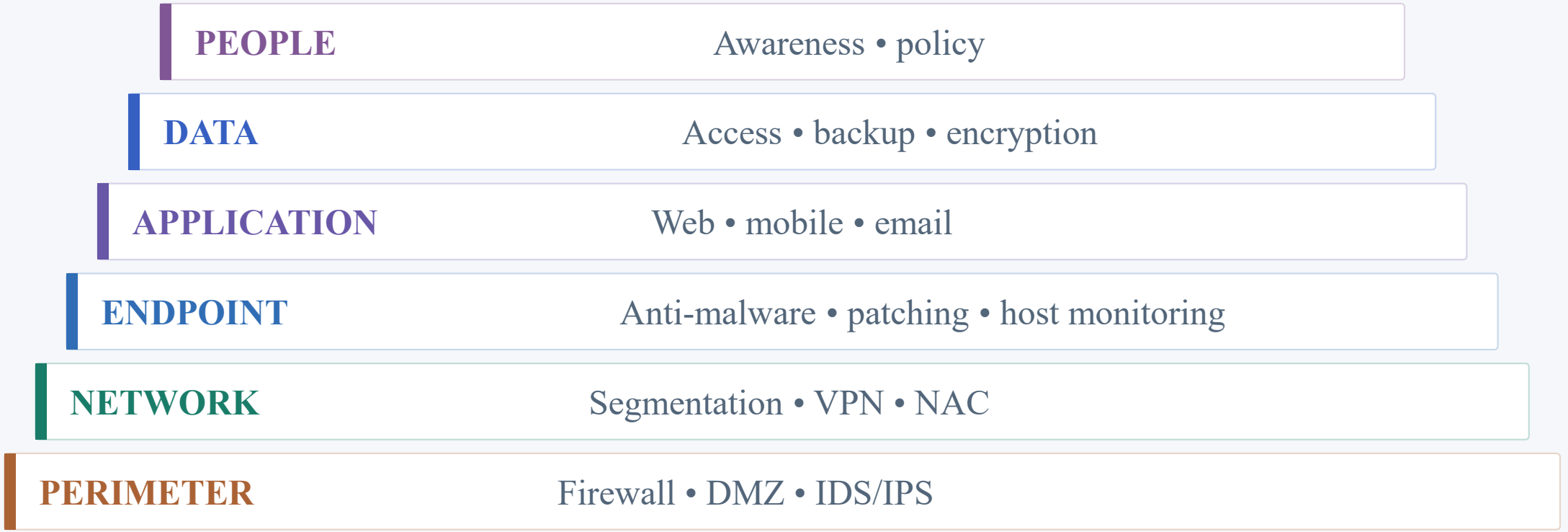
DEFENSE IN DEPTH

Integrate all layers into one security strategy.



Defense in Depth stack

Every layer protects a different part of the system.



Cloud and workload security span multiple layers rather than sitting outside the model.

Defense in Depth is more than prevention

A robust strategy plans for the full security lifecycle.



PREVENT

Reduce chance of compromise



DETECT

Notice unusual activity



RESPOND

Contain and manage



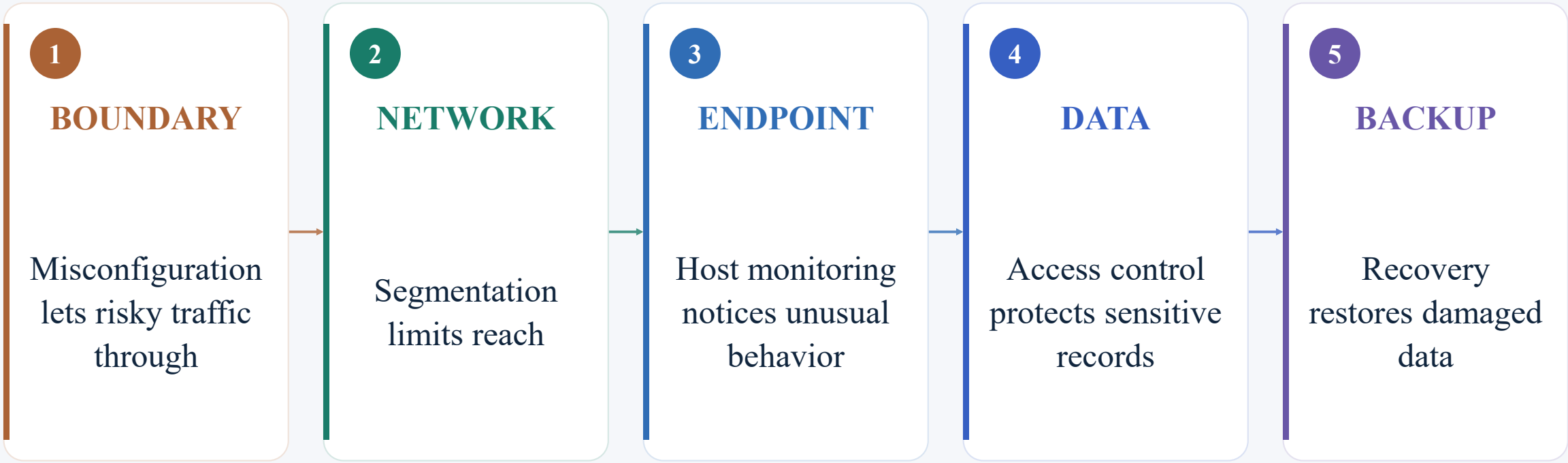
RECOVER

Restore service and data

> Incident response and reporting belong to Defense in Depth too.

If one layer fails…

The next layer should still have a chance to reduce harm.



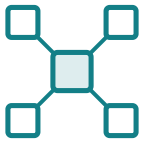
Layered security creates multiple opportunities to detect, contain, and recover.

Benefits of a layered security approach



REDUCED RISK

Less dependence on one control.



COMPREHENSIVE PROTECTION

Different layers address different problems.



EARLY DETECTION

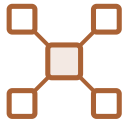
Later controls may catch what earlier controls miss.



COMPLIANCE SUPPORT

Layered controls can support regulatory expectations.

Challenges of layered security



COMPLEXITY

More tools and teams to manage.



FALSE CONFIDENCE

Poorly maintained layers still leave gaps.



USER FRICTION

Security steps can inconvenience users.



INTEGRATION

Controls may not work smoothly together.



COST

Hardware, software, and skilled staff require investment.

Implementation best practices

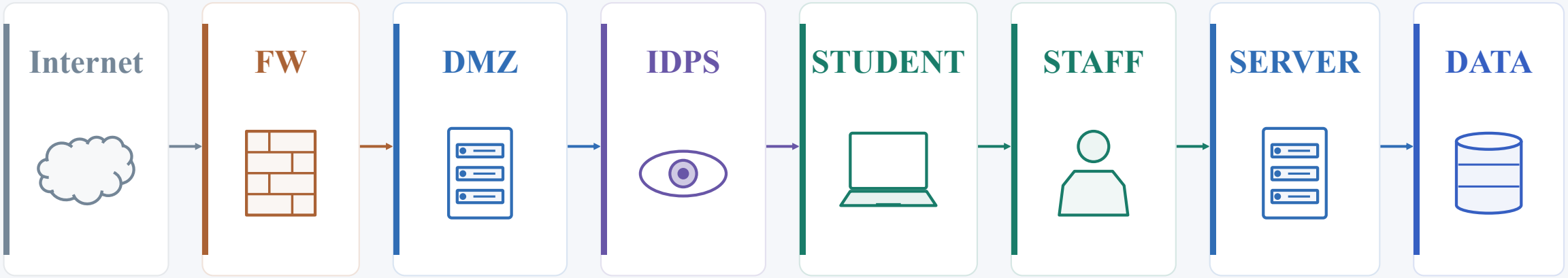
Security posture improves through continuous work.



No layer stays effective without maintenance, people, and measurement.

Complete university security architecture

Map the controls to the layers they protect.



VPN / NAC | ENDPOINT | APP SECURITY | DATA SECURITY | USER / POLICY | CLOUD / IAM

> Defense in Depth = coordinated controls across every layer.

Week 8 visual summary

Security is a system of coordinated layers — not a single product.

PERIMETER

First boundary

NETWORK

Control internal movement

ENDPOINT

Protect devices

APPLICATION

Protect software behavior

DATA

Control, encrypt, recover

USER

Awareness and policy

CLOUD

Protect distributed environments

DEFENSE IN DEPTH

Integrate all layers

> This week: understand the defenses.

Next week: Cyber Attacks

After understanding the defenses, we study how attackers try to bypass them.

THIS WEEK

Security layers
Defense in Depth
Where controls belong
How controls work together

NEXT WEEK

Threat actors
Attack vectors
Social engineering
Malware & network attacks
Attack lifecycle

> Learning order matters: know what is protected before studying how it is attacked.

Exit question

Use the complete layered-security model.

A university has a firewall and secure Wi-Fi. What other layers are still needed to protect the whole information system?

ENDPOINT | APPLICATION | DATA | USER | CLOUD

DEFENSE IN DEPTH